

AWS Cloud Infrastructure Assignment

S3 • IAM • EC2 • VPC Networking

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AWS Region: us-east-1

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1. Objective

The objective of this assignment is to create basic AWS cloud infrastructure, configure secure storage and IAM access, launch EC2 instances, configure VPC networking, and demonstrate secure access using AWS services.

- Amazon S3 private object storage
- IAM custom policy and IAM role
- Amazon EC2
- Amazon VPC and subnet configuration
- Internet Gateway and route tables
- Security Groups
- Public EC2 deployment and HTTP verification

2. AWS Services Used

AWS Service	Purpose
Amazon S3	Private object storage
IAM	Identity and access management
Amazon EC2	Virtual server
Amazon VPC	Network isolation
Subnets	Network segmentation
Internet Gateway	Internet connectivity
Route Tables	Traffic routing
Security Groups	Instance-level firewall

Task 1 — Create S3 Storage

3. S3 Bucket Creation

An S3 bucket named student-cloud-kiruthika-2026 was created and configured as a private bucket.

Configuration	Value
Bucket Name	student-cloud-kiruthika-2026
Region	us-east-1
Access	Private
Public Access	Blocked

Screenshot 1 — Insert screenshot of the S3 bucket showing the bucket name and configuration.

4. S3 Object Upload

The required objects were uploaded to the bucket:

- cloud.jpg
- student-guide.pdf
- notes.txt

Screenshot 2 — Insert screenshot of the S3 Objects page showing cloud.jpg, student-guide.pdf, and notes.txt.

5. Private S3 Access

The bucket was kept private by maintaining Block Public Access. No public bucket access was configured. Authorized AWS identities, such as the EC2 instance through its IAM role, can access the required objects.

The S3 website endpoint returned 403 AccessDenied because the bucket is private. This is expected for a private S3 bucket.

Screenshot 3 — Insert screenshot of S3 Permissions showing Block Public Access enabled.

6. S3 Object Download

The EC2 instance successfully downloaded an object from the private S3 bucket using the attached IAM role.

Command: `aws s3 cp s3://student-cloud-kiruthika-2026/student-guide.pdf .`

Task 2 — IAM Policy and Role

7. Custom IAM Policy

A custom IAM policy was created with only the permissions required for the assignment: `s3:ListBucket` and `s3:GetObject`. The resources are restricted to the student's own S3 bucket.

- No `s3:DeleteObject` permission
- No unrestricted S3 access
- No IAM administration permissions

This follows the principle of least privilege.

8. IAM Role

An IAM role named `StudentCloudEC2Role` was created and attached to the EC2 instance. The EC2 instance obtains temporary AWS credentials through the IAM role. No AWS access keys were stored on the EC2 instance.

Architecture: EC2 Instance → `StudentCloudEC2Role` → Custom S3 IAM Policy → `student-cloud-kiruthika-2026`

Screenshot 4 — Insert screenshot of IAM Role showing `StudentCloudEC2Role` and its attached policy.

9. Verify IAM Role from EC2

The following command confirmed that the EC2 instance was operating using the `StudentCloudEC2Role`:

```
aws sts get-caller-identity
```

The result showed an assumed-role ARN similar to: `arn:aws:sts::981026211544:assumed-role/StudentCloudEC2Role/i-0a07daa5308b61e77`

Screenshot 5 — Insert screenshot of the EC2 terminal showing `aws sts get-caller-identity`.

10. Test S3 List Permission

The following command successfully displayed the objects in the bucket:

```
aws s3 ls s3://student-cloud-kiruthika-2026
```

This demonstrates that the IAM role has s3:ListBucket permission.

Screenshot 6 — Insert screenshot of the EC2 terminal showing successful aws s3 ls.

11. Test S3 GetObject Permission

An S3 object was successfully downloaded using:

```
aws s3 cp s3://student-cloud-kiruthika-2026/student-guide.pdf .
```

This demonstrates that the IAM role has s3:GetObject permission.

12. Test Least Privilege

The following command was executed:

```
aws s3 rm s3://student-cloud-kiruthika-2026/student-guide.pdf
```

The command returned AccessDenied because the IAM policy does not allow s3:DeleteObject. This demonstrates least-privilege access.

Operation	Result
List S3 bucket	Allowed
Download S3 object	Allowed
Delete S3 object	AccessDenied — Expected

Screenshot 7 — Insert screenshot of the EC2 terminal showing AccessDenied for aws s3 rm.

Task 3 — EC2

13. EC2 Instance

An Amazon Linux EC2 instance was configured for the AWS lab and associated with StudentCloudEC2Role. AWS CLI was used from the EC2 instance to access the private S3 bucket.

Parameter	Value
Operating System	Amazon Linux
IAM Role	StudentCloudEC2Role
S3 Access	IAM Role
AWS Access Keys	Not configured
S3 Bucket	student-cloud-kiruthika-2026

Screenshot 8 — Insert screenshot of EC2 instance details showing instance, IP addresses, and IAM role.

Task 4 — AWS Networking

14. VPC Creation

A custom VPC named student-cloud-vpc was created with CIDR 10.0.0.0/16.

The VPC provides an isolated virtual network for AWS resources.

Screenshot 9 — Insert screenshot of VPC showing student-cloud-vpc and 10.0.0.0/16.

15. Public Subnet

A public subnet named student-public-subnet was created with CIDR 10.0.1.0/24. The subnet is associated with a route table containing a route to the Internet Gateway.

Screenshot 10 — Insert screenshot of the public subnet showing 10.0.1.0/24.

16. Private Subnet

A private subnet named student-private-subnet was created with CIDR 10.0.2.0/24. The private subnet does not have a direct route to the Internet Gateway.

Screenshot 11 — Insert screenshot of the private subnet showing 10.0.2.0/24.

17. Internet Gateway

An Internet Gateway named student-cloud-igw was created and attached to student-cloud-vpc. It provides Internet connectivity for resources in appropriately routed public subnets.

Screenshot 12 — Insert screenshot of the Internet Gateway showing Attached state.

18. Public Route Table

A route table named student-public-rt was created and associated with the public subnet.

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	Internet Gateway

This configuration allows resources in the public subnet to communicate with the Internet when they have a public IP and appropriate Security Group rules.

Screenshot 13 — Insert screenshot of the public route table showing 0.0.0.0/0 → Internet Gateway.

19. Private Route Table

A separate route table named student-private-rt was created for the private subnet. It contains the local VPC route and no direct Internet Gateway route.

Screenshot 14 — Insert screenshot of the private route table and subnet association.

20. Security Group

A Security Group named student-cloud-sg was created for the public EC2 instance.

Type	Protocol	Port	Source
SSH	TCP	22	My IP
HTTP	TCP	80	0.0.0.0/0

SSH access was restricted to the required source. HTTP was allowed for the public web-server demonstration.

Screenshot 15 — Insert screenshot of the Security Group showing SSH and HTTP rules.

21. Public EC2 Instance

A second EC2 instance named student-public-ec2 was launched in the new VPC to demonstrate public-subnet networking. The existing EC2 instance in the default VPC was left unchanged and continued to be used for the S3/IAM demonstration.

Parameter	Configuration
VPC	student-cloud-vpc
Subnet	student-public-subnet
Subnet CIDR	10.0.1.0/24
Public IP	Enabled
Security Group	student-cloud-sg

Screenshot 16 — Insert screenshot of student-public-ec2 showing VPC, subnet, public IP, and private IP.

22. EC2 Instance Connect

The public EC2 instance was accessed using EC2 Instance Connect directly from the AWS Console. No local PowerShell or local SSH client was required.

Screenshot 17 — Insert screenshot of the EC2 Instance Connect terminal.

23. HTTP Server Configuration

Apache HTTP Server was installed on the public EC2 instance.

```
sudo dnf install -y httpd
```

```
sudo systemctl enable --now httpd
```

```
echo "Hello from student-public-ec2" | sudo tee /var/www/html/index.html
```

24. HTTP Access Verification

The public IP address of student-public-ec2 was opened in a web browser. The page successfully displayed: Hello from student-public-ec2.

This confirms that the EC2 instance has a public IP, is located in the public subnet, the public route table has an Internet Gateway route, the Security Group permits HTTP traffic on port 80, and Apache is running.

Screenshot 18 — Insert screenshot of the browser displaying “Hello from student-public-ec2”.

25. Final AWS Architecture

Internet → Internet Gateway → Public Route Table → Public Subnet (10.0.1.0/24) → student-public-ec2 → Security Group → Apache → HTTP response.

The VPC is 10.0.0.0/16 and also contains the private subnet 10.0.2.0/24 with a private route table. Separately, the S3/IAM demonstration uses EC2 → StudentCloudEC2Role → Custom IAM Policy → private S3 bucket.

26. Security and Least Privilege

- The S3 bucket was kept private.
- Public access to the S3 bucket was blocked.
- The EC2 instance accessed S3 through an IAM role.
- AWS access keys were not stored on the EC2 instance.
- The IAM role was restricted to the required S3 bucket.
- Only ListBucket and GetObject permissions were granted.
- Delete access was intentionally not granted.

- The aws s3 rm operation returned AccessDenied.
- SSH access was restricted through the Security Group.
- Public HTTP access was enabled only for the networking demonstration.

27. Final Results

Requirement	Status
Create S3 bucket	Completed
Upload cloud.jpg	Completed
Upload student-guide.pdf	Completed
Upload notes.txt	Completed
Configure S3 as private	Completed
Download S3 object	Completed
Create custom IAM policy	Completed
Create IAM role	Completed
Attach IAM role to EC2	Completed
Avoid AWS access keys on EC2	Completed
s3:ListBucket test	Successful
s3:GetObject test	Successful
Delete test	AccessDenied as expected
Create VPC	Completed
Create public subnet	Completed
Create private subnet	Completed
Create Internet Gateway	Completed
Configure route tables	Completed
Create Security Group	Completed
Launch public EC2	Completed
EC2 Instance Connect	Successful
HTTP test	Successful
Browser test	Hello from student-public-ec2

28. Conclusion

The AWS assignment was successfully completed by creating a private S3 storage environment, implementing IAM-based least-privilege access, configuring Amazon Linux EC2 instances, and creating a custom VPC with public and private subnets. The EC2 instance accessed the private S3 bucket through an IAM role without using AWS access keys. The S3 delete operation was denied as expected, demonstrating least privilege.

A separate EC2 instance was launched in the custom public subnet to demonstrate AWS networking. The Internet Gateway, route table, subnet, and Security Group were configured correctly, and an Apache web server was successfully accessed through the instance's public IP address. The successful display of “Hello from student-public-ec2” confirmed public HTTP connectivity.